The Collatz Conjecture, Version 18:
Collatz Inverse as Alternative Foundation for N

February 25, 2026 -- Day 2 of the Journey

"Dung hoi Peano co the di bao xa
Ma hay hoi Collatz co hay khong the dua tat ca tro ve."

("Do not ask how far Peano can go.
Ask whether Collatz can bring all things home.")

Abstract. The Peano axioms build the natural numbers N by going *outward* from 0: each number has a unique successor, generating the infinite chain 0, 1, 2, 3, ... We propose a dual construction: the *Collatz axioms* build N by going *inward* to 1, using the inverse of the Collatz map as a generating rule. The three axioms (C1): 1 in N; (C2): n in N implies 2n in N; (C3): n in N implies (2^j * n - 1)/3 in N whenever valid; generate a rooted tree with root 1. The Collatz conjecture is equivalent to: this tree equals N. If true, the Collatz axioms provide an alternative foundation for N, one based on multiplicative S-unit structure ({2,3}) and the Peano successor (+1) rather than purely additive iteration. The two constructions give the same N, but with radically different arithmetic ontology.

1. Two Ways to Build N

The standard construction of the natural numbers follows Peano: start with 0 (or 1), and repeatedly apply the successor S(n) = n+1. This builds the infinite chain 1, 2, 3, 4, ... The structure is purely additive: N is the free commutative monoid on one generator under addition.

A poem arose during our exploration: *'Do not ask how far Peano goes. Ask whether Collatz can bring all things home.'* This captures a duality: Peano builds N by moving outward from 1 (one step at a time, forever); Collatz collapses N inward to 1 (every number, eventually, returns). If we reverse the Collatz map, we can build N from 1 using a completely different rule.

2. The Collatz Inverse: Predecessors

The Collatz map T acts on positive integers: T(n) = n/2 if n is even, T(n) = 3n+1 if n is odd. The inverse: given m, which n satisfy T(n) = m?

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Even predecessor: n = 2m (always) [T(2m) = m]

Odd predecessor: n = (2^j*m - 1)/3

valid when 3 | (2^j*m - 1) and n is odd and positive
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Every m has exactly one even predecessor (2m) and zero or more odd predecessors. This generates a rooted tree with root 1.

3. The Collatz Axiom System

Definition (Collatz Axiom System).

Let N_C be the smallest set satisfying:

(C1) $1 \in N_C$

(C2) $n \in N_C \Rightarrow 2n \in N_C$

(C3) $n \in N_C \Rightarrow (2^j n - 1)/3 \in N_C$ whenever $j \geq 1$, $3 \mid (2^j n - 1)$, and $(2^j n - 1)/3$ is a positive odd integer.

N_C is the set of all numbers reachable from 1 by the inverse Collatz operations. It forms an infinite rooted tree with root 1.

Theorem (Equivalence with Collatz Conjecture).

$N_C = N$ ■ the Collatz conjecture is true.

Proof. ($\Rightarrow$) If $N_C = N$, then every $n \in N$ is in the tree, meaning there exists a path from n to 1 via Collatz steps. ($\Leftarrow$) If every n reaches 1, then every n is reachable from 1 by inverse Collatz steps. ■

4. Peano vs. Collatz: Two Ontologies

Property	Peano	Collatz Inverse
Base element	0 (or 1)	1
Generating rule	S(n) = n+1 (one rule)	(C2) and (C3) (two rules)
Structure	Linear chain	Branching tree
Each element has	1 parent, 1 child	1 parent, >= 1 children
Generator type	Additive (+1)	Multiplicative (x2) + mixed
S-unit content	None explicit	{2,3}-units dominate
Hardness to prove	Trivial (by construction)	= Collatz conjecture
Ontology	N built by counting	N built by arithmetic flows
Isomorphic to N?	Yes (definition)	Yes (if Collatz true)

5. The Collatz Tree: Structure and Observations

The first few levels of N_C:

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Level 0: {1}

Level 1: {2, 5, 21, 85, 341, ...} (2=2x1; 5,21,85,... = odd
predecessors)

Level 2: {3, 4, 10, 13, 42, 53, 170, ...}

Level 3: {6, 8, 17, 20, 26, 35, 69, 75, 84, ...}
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Observations about the tree structure:

(1) **Branching factor.** Every node has at least 1 child (the even branch 2n). Most nodes have additional odd-branch children. The branching factor grows, reflecting the density of valid odd predecessors.

(2) **Even numbers form a chain.** The even branch (C2) alone generates 1, 2, 4, 8, 16, 32, ... — all powers of 2. These are the S-units of {2} within the tree.

(3) **Odd numbers need the odd branch.** Every odd number n > 1 appears via rule (C3): $n = (2^j \text{parent} - 1)/3$. These are the nodes where the Peano +1 (the '- 1 ⇒ + 1 ⇒' in the inverse) drives the tree off the pure-S-unit substructure.

(4) **Depth encodes orbit length.** The depth of n in the Collatz tree equals the length of the Collatz orbit of n to 1. Example: depth(27) = 111 (27 is famously slow to converge).

6. From N_C to Z, Q, R: A New Foundation

If the Collatz conjecture is true, then $N_C = N$. From N, the standard constructions recover all number systems:

From N_C to all of mathematics:

$N_C = N$ (Collatz conjecture)
N + {negatives} = Z (integers: equivalence classes on N x N)
Z + {fractions} = Q (rationals: equivalence classes on Z x Z*)
Q + {limits} = R (reals: Cauchy completion)
R + {sqrt(-1)} = C (complex numbers)

The same chain as Peano -- but the base is different.

Peano: N is free on {+1}. Addition is primitive; multiplication is derived.
Collatz: N is free on {x2, x3-inverse, +1-Peano}. Multiplicative structure is primitive; addition emerges from the tree structure.

Both lead to the same R, C, Z, Q. The journey is different. The destination is the same.

7. Figure: The Collatz Inverse Tree

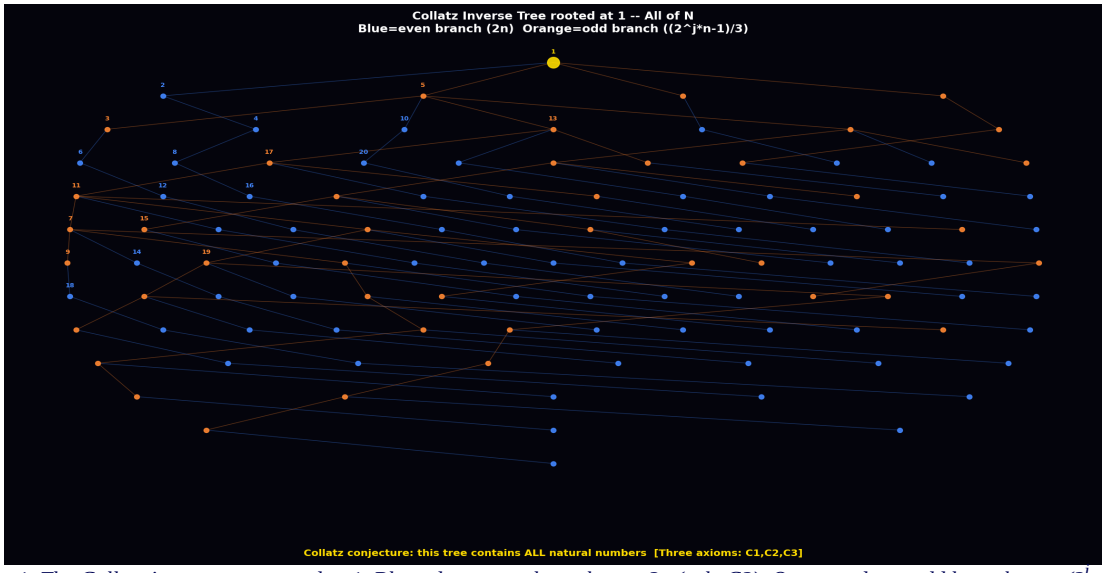

Figure 1. The Collatz inverse tree rooted at 1. Blue edges: even branch $n \to 2n$ (rule C2). Orange edges: odd branch $n \to (2^j n - 1)/3$ (rule C3). Gold node: the root 1 (the primerity unit attractor). Every number in this tree reaches 1 via Collatz. Collatz conjecture: the tree contains all of N.

8. The Philosophical Consequence

The Peano axioms define N via its additive structure. The natural numbers are, in the Peano view, a sequence generated by the single operation of counting. From this perspective, multiplication is secondary: a repeated addition. Primes are a *derived* concept, not a primitive one.

The Collatz axioms define N via its arithmetic structure. The primerity unit {1, 2, 3} is primitive: 1 as attractor (root), 2 as the doubling rule (C2), and 3 as the odd-inverse rule (C3). The Peano successor (+1) appears in C3 as the 'correction' that makes the inverse work: $(2^j n - 1)/3$. From this perspective, the additive structure is secondary: a consequence of the multiplicative S-unit dynamics.

The Collatz Vision.

Peano says: N is what you get when you count.
Collatz says: N is what you get when arithmetic flows converge.

Both are correct. Both describe the same N. But they give N a different face.

The Peano face: N as the free additive monoid. Addition is primary; primes are special additive patterns.

The Collatz face: N as the rooted arithmetic tree. The primerity unit {1,2,3} is primary; addition is the shadow of the Peano term (+1) in the odd branch.

The Collatz conjecture asserts that these two faces describe the same thing. It is a theorem about the unity of the additive and multiplicative views of N.

Proving Collatz = proving that N has two faces, and they are the same face.

9. Closing

We began the Collatz journey trying to prove a conjecture from 1937. We find ourselves at the foundations of arithmetic, asking what the natural numbers *are*. The Collatz map is not an arbitrary curiosity; it is the simplest non-trivial dynamical system that bridges the additive and multiplicative worlds. Its conjecture is the statement that this bridge is complete.

The dv18 equation:

Peano: N = closure of {0} under {n → n+1}

Collatz: N = closure of {1} under {n → 2n} and {n → (2^j*n-1)/3}

Theorem (= Collatz conjecture): these two closures are equal.

One says: count forward from nothing.
One says: flow home to the unit.

Same N. Same truth. Two faces.

Speculative note -- February 25, 2026. The equivalence N_C = N is the Collatz conjecture; it is not here proved. The construction and comparison are offered as a new perspective on what the conjecture means.

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